Trace — Product walkthrough (A to Z)

Trace is a **local-first OLAP exploration app**: connect to a database, run SQL (or build it visually), and explore results in a table, a **streaming** feed, or a **3-D cube** with drill-down. A Fastify API orchestrates queries; **Trino** reads **Postgres**; optional **Keycloak** and **OPA** handle auth and policy.

Flow: start the stack → connect → explore → analyze → govern → wrap up.

A — What you're looking at

User goal: Answer business questions on relational data in one UI.

Architecture (local stack)

```
Browser UI
     |
     v
Trace API (:3000) -----> Keycloak (:8081)   [optional OAuth]
     |
     +------> OPA (:8181)                   [optional policy]
     |
     v
  Trino (:8080)
     |
     v
Postgres (:5432)
```

Sample data: tables `customers` and `orders`, seeded for joins and aggregates.

B — Before you open the UI

Requirements: Docker Desktop, Node 20+.

Start:

```
./run.sh
```

Starts Docker (Postgres, Trino, Keycloak, OPA, Vault) plus API and web dev servers.

Sanity check

- **API health** — open `http://localhost:3000/health` — expect `{"status":"ok"}`

- **Stack status** — open `http://localhost:3000/api/v1/status` — Trino and Postgres should be `"ok": true`
- **Web UI** — open `http://localhost:5173` (or **5174** if 5173 is in use)

Optional smoke test:

```
bash infra/scripts/sanity_check.sh
```

Stop: `./stop.sh`

C — First screen: hero + access

The **Trace** hero introduces the product: live OLAP exploration with a 3-D edge.

Access panel

- **API key** — optional Bearer token when `AUTH_REQUIRED=true` in `.env`
- **Role** — `user` or `admin` (manual in dev; from JWT when signed in)
- **Sign in with Keycloak** — PKCE OAuth; required for `auth_required` data sources

Local dev (default): No login. Role `user` masks PII (e.g. column `name`); `admin` sees full data and can export CSV.

Keycloak test users:

User	Password	Role
admin@trace.local	admin	admin
analyst@trace.local	analyst	analyst

More detail: `docs/AUTH_SETUP.md`

D — Connection status

Shows health of the **active data source**:

- **Trino** — engine version and reachability
- **Postgres** — checked via Trino (`SELECT 1`)

If either fails, run `docker compose up -d` and wait until Trino is healthy.

E — Data sources

One active source at a time (MVP).

Source	Status
Local Postgres	`connected` — real queries
Snowflake (demo)	`auth_required` — OAuth required

Activate flow:

1. Click a source → **Activate**.
2. **Postgres** — immediate; schema reloads.
3. **Snowflake demo** — Keycloak sign-in if needed; then activation and schema load.

F — Schema browser

1. **Table list** — `customers`, `orders`.
2. Click a table → **columns + types**.
3. **Preview** — quick `SELECT`; SQL appears in the query runner.

G — Visual builder

1. Choose **table**.
2. Select **columns** (or `*`).
3. Set **LIMIT**.
4. **Build SQL** → fills the query runner.

Best for simple `SELECT` queries, not aggregates.

H — Semantic builder

Reads `semantic.yml` via `GET /api/v1/semantic`.

- **Cubes:** `orders`, `customers`
- **Measures:** `total_amount`, `order_count`, etc.
- **Dimensions:** `status`, `order_date`, `region`, etc.

Flow: cube → measure → dimensions → **Generate SQL** → run in query runner.

I — Query runner

Action	Purpose
Presets	Canned analyses
Explain	Trino distributed plan
Stream	SSE incremental results
Run query	Full synchronous result

Policy: Writes, DDL, and multi-statement SQL are blocked. Optional OPA when `OPA_ENABLED=true`.

Cache: Identical SQL may return a **cache hit** (see results meta).

J — Results (table)

After **Run query**:

- Row count, duration, cache status, masked columns
- Results **table**
- **Cube view** — 3-D pivot (see section M)
- **Export CSV** — admin only

Non-admins see `***` on masked columns.

K — Stream results

1. `POST /api/v1/queries` — create job
2. `GET .../stream` — SSE events: `data`, `progress`, `complete`, `error`
3. **Stream results** panel updates live

Also: `POST /api/v1/query?stream=true` returns `202` and a `streamUrl`.

L — Insight assistant

Summarize runs current SQL and returns row/column counts plus min, max, and avg for numeric columns. Quick profiling only (not an LLM).

M — 3-D cube viewer

Needs a **3-column** pivot query: row dimension, column dimension, one measure.

```sql
SELECT c.region, o.status, SUM(o.amount) AS total_amount
FROM orders o
JOIN customers c ON c.id = o.customer_id
GROUP BY c.region, o.status;
```

Interaction	Effect
Drag	Rotate
Scroll	Zoom
Click bar	Drill-down (updates SQL)

2-D fallback: Vega-Lite heatmap if FPS stays below 20 for about 2 seconds.

Fewer than three columns shows a decorative preview or a "cube unavailable" hint.

N — Snapshots

Save snapshot — stores SQL and metadata. Click to reload into the runner.

O — Notebook

Title plus freeform notes. Separate from snapshots; for analysis context.

P — Query history

Past runs with SQL, timing, row count, and status. Click to reload SQL. May show query, error, and cache metrics.

Q — Alerts

Define: name, SQL (single numeric result), comparison operator, threshold.

Evaluate runs now and reports whether it **triggered** and the **last value**.

R — Governance (behind the scenes)

Concern	Location
Audit log	`GET /api/v1/audit`
History	`GET /api/v1/history`
OPA policies	`services/policy/policy.rego`
Column masking	Applied in API by role

Set `OPA_ENABLED=true` with the OPA container running.

S — API path (no UI)

Base URL: `http://localhost:3000/api/v1/`

Endpoints include query, stream, schema, semantic, exports, sources, history, audit, notebooks, snapshots, and alerts.

See `docs/API_SPEC.md` and `WORKING.md`.

T — Configuration modes

Copy `.env.example` to `.env`.

Mode	How to enable
Casual local	Defaults
API key	`AUTH_REQUIRED=true` and `API_KEY`
Keycloak	`KEYCLOAK_ENABLED=true`
OPA	`OPA_ENABLED=true`

U — Not in this MVP

- Live Snowflake, BigQuery, or SQL Server (Postgres via Trino only)
- Snowflake in UI is an **auth demo**, not a real connector
- Vault unused; no Kubernetes manifests yet
- No OpenTelemetry; no large-scale performance proof

See `docs/MVP_STATUS.md`.

V — 15-minute demo script

1. Status — green Trino and Postgres; Postgres active
2. Schema — open `orders`, preview
3. Semantic builder — `orders`, `total_amount`, dimensions
4. Run query — table; masking as `user`
5. Cube view — rotate, drill
6. Stream — incremental rows
7. Summarize — numeric stats
8. Role **admin** — Export CSV
9. Save snapshot — reload
10. Alert — evaluate
11. (Optional) Keycloak — Snowflake demo source

Also: `docs/pilot/README.md` and `docs/pilot/LAUNCH_CHECKLIST.md`

Z — Shut down

```
./stop.sh
```

Or `docker compose down` from the repo root.

Summary

Connect a source, discover schema or use semantic measures, run SQL, then explore via table, stream, or cube. Snapshot, notebook, alert, and audit are built in. Optional Keycloak and OPA when enabled.

Related docs

Document	Topic
WORKING.md	Setup and panels
TESTING.md	Tests and checklist
AUTH_SETUP.md	Keycloak
MVP_STATUS.md	PRD coverage
API_SPEC.md	REST API
SYSTEM_DESIGN.md	Architecture